NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.9 Nature Review Studio Review Coverage Quick Trial

Group Rapid Trials Lab

Status Closed (negative result)

Period 2026-10-19 22:26 — 2026-10-26 12:17

Stage conclusion

Project report

L26.9 Nature Review Studio: review coverage quick trial

Launch plan: (submitted 10-19 22:40) Gate: **cleared**; the platform approved L26.9 on 10-19, based on the PI's own words at 10-19 22:4x. Deliverable destination: the T34.2 "Paper Review" lead.

Latest: NHESS subsample results (10-26 12:2x, approved by the PI; closure proposed, awaiting PI review)

Component	Criterion (set 10-19, unchanged)	NHESS subsample result	Comparison with 10-21 primary sample	Verdict
12-axis checklist	Δ coverage ≥10 percentage points	A 27.3% / B 29.2%, Δ +1.9, 95% CI [−1.0, +4.7] (10 papers, 314 units); major comments only Δ +3.6 [−2.4, +9.5]; including partial coverage Δ −0.1	Primary sample Δ +0.1 [−2.2, +2.4]; pooled 20 papers, 747 units Δ +0.8 [−0.9, +2.6]	**Abandon (no practical gain), same direction**
Response-action dictionary	Annotatable ≥85% and κ≥0.6	78.7% (211/268), κ 0.73	Primary sample 84.3%; pooled 80.1% (286/357)	**Abandon, same direction**
manuscript drafting	Aligned rate ≥60% in dictionary group	Not re-run on NHESS (see "Decision I made" below)	Primary sample 70% (with dictionary) / 75% (without)	Retain primary sample conclusion

- Judging reliability: Pairwise κ across two judges 0.80–0.98; 119 disagreements blindly adjudicated by the lead; across all four readings (intersection, union, and each individual judge), Δ falls between +1.8 and +2.3, with upper interval bounds all ≤ +5.0. Invalid model comment items were 0 in both groups.
- At the single-paper level, only H8 (insurance loss model and meteorological loss index, 62 units) showed Group B higher by 11.8 pp; the remaining 9 papers ranged from −3.7 to +6.7. Single-paper variations are not attributed.
- Positioning and limitations of NHESS: A geoscience/natural hazards journal with an open discussion peer-review model (RC/AC fully public, CC BY) belonging to the same Copernicus system as WES, so the two subsamples share a similar peer-review culture and do not represent a truly independent "different class of journal"; the 10 papers focus primarily on European windstorm losses, tropical cyclones, convective windstorms, tornado climatology, and insurance/risk, with **not a single structural wind engineering paper (wind loads, wind-induced vibration, component fragility)**—NHESS rarely publishes such papers, and none appeared among the 229 retrieved candidate titles; the legacy discussion system prior to 2016 is unavailable, so the actual publication years span 2017–2025; strata N7 (forest windthrow) and N10 (infrastructure) had no eligible candidates, so 2 papers were drawn from the pooled reserve per the 11:45 supplementary rule (the 8-paper reading excluding these 2 papers gives Δ +2.3 [−0.8, +5.6], leaving the conclusion unchanged); author responses for H7 were inside a zip attachment and not extracted, reducing the dictionary sample by 44 units.
- Scope limitations of conclusion: These two samples (18 papers across two Copernicus journals + 2 Nat Commun papers), this model (generation and judging within the same Claude model family), and these prompts. Does not extrapolate to structural wind engineering journals or other models.
- **Decision I made**: Did not re-run the manuscript drafting test on the NHESS subsample—the dictionary already failed the annotation threshold twice, and using the dictionary yielded no drafting gain in the

primary sample, so re-running 40 drafted responses would not change the "do not adopt dictionary" verdict—cost if wrong: if the dictionary helps response drafting specifically on wind hazard papers, this was not detected here; if needed, it can be supplemented using the 268 already-split author responses, requiring ~40 manuscript drafting runs.

- Evidence: , ; checksums , . Raw texts, review comments, and detailed judging records are stored only in local non-public directories.

Conclusion (10-21 15:3x, primary sample, per pre-launch criteria)

Component	Criterion	Result	Verdict
12-axis checklist	Δ coverage ≥10 percentage points	A 26.5% / B 26.6%, Δ +0.1, 95% CI [−2.2, +2.4] (433 units); across 10 papers, B higher in 4, B lower in 6	**Abandon (no gain)**
Response-action dictionary	Annotatable ≥85% and κ≥0.6	84.3% (75/89), κ 0.74	**Abandon (missed threshold by 1 item)**; missing 4 action types—adding them might pass the threshold, but the next row shows the dictionary yields no gain for drafting
manuscript aligned response drafting	Aligned rate ≥60% in dictionary group	With dictionary 70% (14/20); without dictionary 75% (15/20); inter-judge agreement 100%	**Passed per criterion**, but gain does not come from the dictionary; only 6–7/12 major comments aligned, with the main flaw being "over-compliance"
"Real-data distillation" selling point	(Verification, no criterion)	Only aggregate counts and checksums are verifiable; classification relies on keyword regex	Unverifiable

Scope limitations of conclusion: This sample (8 WES papers + 2 Nat Commun papers, mostly wind energy/atmospheric/fluids, few structural wind-load papers), this model (generation and judging within the same Claude model family), and these prompts. The "invalid comment" metric was 0 in both groups; judges were lenient, so this metric is not used as a basis. Recommended disposition: Do not integrate the 12-axis checklist into , and do not integrate the dictionary into manuscript; pass only the single lesson—"when drafting responses, remind that authors may have valid reasons not to comply, so first judge whether the requested change should be made"—to the T34.2 "Paper Review" lead for reference. Falls under "Exit X2: keep format only".

Process status (10-21 14:4x)

- **Reference source changed to b (decided by the PI on 10-21)**: 8 open-peer-review papers from Wind Energy Science + Nature Communications P1, P2 (reviewing original submitted versions). The PI's words: "Okay, proceed with everything as you suggested. Note that if literature is needed, it's best to search for literature first. Since this is a quick trial, try not to use deep research." Status reverted to Exploring.
- Sampling rules finalized on 10-21 14:42: ; comment splitting rules finalized at 14:43; see for frozen file sha256 hashes.
- Batch 1 (P1, P2): Ground-truth comment units frozen at 35 for P1 (sha256 …) and 54 for P2 (…); 12 model review reports generated; blinding key table sha256 23c0b3e57bde8f7285f97dc35cc093a1250231e94285b5ea82d879180f9b5a7f (recorded prior to judging).
- Batch 2 (W1–W5) blinding key table sha256 dad3095cc166aeea10d54ca4f4bf01ed6f5e7b8d51dff47fe3a5c3f5065b8f81 (recorded prior to judging on 10-21).
- Batch 3 (W6–W8) blinding key table sha256 9d935fb9262734a8f39225178037f81630de1717101e19c75b7ec33a2da5b31f (recorded prior to judging; blinding script fixed before judging on 10-21: bullet points inside two review reports had been mistakenly split into standalone items and were merged back into their parent numbered items before re-blinding; old key table voided and unused).
- **Deviations (recorded faithfully)**: (1) Response-action annotation was completed by two independent sub-agents rather than the lead personally annotating one set as planned (the lead only verified statistics and unannotatable items); (2) The plan stated "open the blinding key table only after all judging is complete", whereas the Batch 1 key table was opened after Batch 1 was judged but before Batch 2 was generated (the lead viewed intermediate P1/P2 results early). Disagreement verdicts for subsequent batches were still performed blind to group assignment, and generation/judging prompts were unchanged.
- Completed (all under "reading materials"):
- Static repository audit: GitHub API and raw text, copies and sha256 hashes stored in , not for publication
- Verified and downloaded PRF links for 3 sample papers to (not for publication)
- Literature search.
- Not done: Any review generation, judging, or bot delegation.

Settled decisions

- Sample: P1 LES of urban cloud enhancement (doi:10.1038/s41467-026-68986-0) and P2 vertical-axis wind turbine pitch control (doi:10.1038/s41467-024-46988-0) serve as primary samples, both reviewing the original Research Square submission versions; P3 offshore wind farm wake (doi:10.1038/s41467-026-68655-2) has no preprint and serves only as a supplement. Pending confirmation by the PI in plan §7.

- Repository text and raw PRF files will not enter public directories; public pages will contain only design, statistics, and conclusions.

Trials and results

Date	Action taken	Result	Evidence
10-21	Checklist coverage (10 papers: 8 WES + 2 Nat Commun; 433 ground-truth comment units; 3 reports per group per paper, 60 model review reports total; dual blind judging + blind adjudication by the lead, κ 0.75–0.94)	Primary reading A 26.5% / B 26.6%, Δ +0.1 percentage points, 95% CI [−2.2, +2.4]; including partial coverage Δ +0.1; major units only Δ −0.5. Per §2.5 criterion: **no gain → abandon checklist**	
10-21	Response-action dictionary annotation (89 author responses across P1 and P2, two independent annotators)	Annotatable 84.3% (75/89), κ 0.74; criterion ≥85% and κ≥0.6 → **missed threshold by 1 item**; missing four action categories: "reply without manuscript revision", "trim main text", "add figure for existing results", and "partial response"	
10-19	Static audit of "distilled from 1287 real PRFs"	Only aggregate counts and source-file checksums are verifiable; corpus is unpublished; 12 axes and severity levels are tagged via keyword regexes, contradicting README claims of "manual categorization" and "≥5% marked major"; "21 response actions" table actually lists 20	Launch plan §4.3; (not for publication)

Open items and next steps

- 10-19 22:55 Completed sample-independent preparations: (self-written checklist and two prompt sets, pending freeze), (synthetic data smoke test passed). The platform has approved the plan.
- Once the PI approves the launch plan → start work per §2.7 milestones; before starting, delegate litwiki to build a card and verify page numbers for Liang et al. 2024 (NEJM AI).
- Minor fix suggestion for close-reading cards (sent to the coordinating lead to forward to the knowledge line): rendering script exists, only the path in SKILL.md is mismatched.

Literature basis

See launch plan §4.2. Card built by litwiki on 10-21: - Liang et al. 2024, NEJM AI, doi:10.1056/AIoa2400196, card. Their "overlap rate" is a directional hit rate with model key points in the denominator (GPT-4→single human: Nature family 30.85%, human→human 28.58%; published PDF p. 4–6), with key points extracted by GPT-4 and matches judged by GPT-4 (similarity ≥7), achieving human spot-check matching F1 of 0.824 (preprint PDF p. 9). **Coverage in this quick trial uses human comment units as the denominator and merges all reviewer comments for the same paper; the definitions differ, so the numbers are not directly comparable.** - Brief cards for comparable precedents: Tyser 2024 (arXiv:2408.10365), Fichtl 2026 (arXiv:2608.03581), in the same directory.

Parameter source table

Parameter	Value	Source / Assumption
Coverage gain threshold	10 percentage points	The PI's original request / close-reading card addendum
Max comments per review report	25	Assumption: prevent Group B from winning by sheer volume; actual ground-truth comment unit counts for P1/P2 to be verified at launch and adjusted if exceeding 25
Replicates per group	3	Assumption: control generation randomness within affordable quota
Dictionary annotatable threshold / κ	85% / 0.6	Assumption: κ 0.6 is the standard lower bound for "substantial agreement"
Aligned drafting rate threshold	60%	Assumption: majority plus margin
Sampling seed	20261019	Convention of this line

Notes from the PI

- 10-19 22:4x, the PI's words: "Let's pick 10 papers, published in JWEIA within the last 20 years, with content and distribution as broad as possible. I agree with the rest, go ahead and execute." → Plan approved; sample changed to 10 JWEIA papers (within 20 years, broad topic distribution).
- 10-21, the PI's approval of item 7 of the L26 summary report: no internal group JWEIA review records → select option b (directory for option a was not provided).
- Original blocker: JWEIA does not publish peer reviews, leaving no reference for coverage. Asked the PI: a) our group's JWEIA manuscript review records / b) switch to Wind Energy Science open peer reviews /

c) change reference standard (not recommended).

Deliverable destination (added by the coordinating lead, 10-19)

- Once the deliverables (trial conclusions on the 12-axis checklist, verification and annotation of the 21 response-action dictionary) are finalized, hand them over via to the T34.2 "Paper Review" lead, without storing a separate copy on this track.

Citation snowballing expansion (10-22, coordinating lead)

- Per the PI's 10-22 request, expanded literature using (OpenAlex 1-hop forward/backward + keywords), screening log.
- New literature not in our library and slated for card creation is listed in the screening log; card creation delegation to litwiki is pending dispatch (platform-wide delegation quota was full at 10-22 00:5x; scheduled callback retry). Conclusions and criteria on this page remain unchanged until cards return; potential impacts are noted in the final section of the screening log.

Relationship with newly added literature (10-22, after cards returned; coordinating lead)

- Conclusions and verdicts remain unchanged. Literature findings that "aspect/topic guidance improves coverage" rely on broader definitions of coverage, which are not the same quantity as this trial's "item-by-item, same-target, same-direction full coverage":
- Yuan, Liu, Neubig 2022 (JAIR): Aspect coverage (ACov) with aspect-assisted training increased by only ~1–1.5 points (50.37→51.50; 62.64→63.96; 55.37→56.91), while aspect recall (ARec) in the summary+CE group actually dropped from 58.31→57.56; the reference was meta-review "aspect + polarity" rather than item-by-item matching (journal p. 175, 184–185, Table 5).
- ReviewEval 2025: topic coverage = proportion of expert topics matched by at least one AI topic reaching a moderate similarity threshold; the authors themselves caution that this cannot be treated as an item-by-item comment hit rate (PDF p. 4 / print p. 20545, Eqs. 1–2).
- Reading: Even under broad definitions, the gain is only on the order of ~1 percentage point, consistent in direction with this trial's "full coverage Δ +0.1 pp [−2.2, +2.4]" and not a counterexample.
- Thakkar et al. 2026 (Nat Mach Intell, ICLR 2025 randomized experiment across 44,831 reviews) addresses "whether reviews are updated after giving human reviewers LLM feedback," finding a mean length increase difference of 14 words in the randomized treatment group—a different question, serving only as background context (preprint PDF p. 3–4, 7–8; for published version see card, PDF p. 2, 4, 6, 8 corresponding to journal p. 327, 329, 331, 333).

10-24 addendum (lead, using existing local materials only)

- Completed run on 10-21 under plan b (Wind Energy Science open peer reviews, 10 papers, 433 first-round review comments), concluding "abandon adoption", header conclusion; see , response_actions_stats.md, and drafting_stats.md for statistics.
- Knowledge base updated: added `Trial: ineffective` to line 20 of source note *.md; method card created (retaining only the "Comment—Evidence—Action—Status" table format).
- Literature added via citation snowballing on 10-22 (Yuan 2022, ReviewEval 2025, Thakkar 2026) does not alter the conclusion; see previous section.
- Current status: No running computations, nothing in transit. Closure proposed (awaiting PI review): a definitive conclusion has been reached per the pre-launch criteria.

10-26 resumption: NHESS subsample (approved by the PI)

- The PI's words on 10-26: "Can we use a journal from the meteorology side? I know they have such journals, for example [external URL omitted] relayed by the coordinating lead with approval: supplement with 10 NHESS open discussion papers (wind hazard priority, within 20 years, diverse topics).
- Protocol unchanged: Criteria (10-19), prompts and checklist, splitting rules, and blind judging workflow (frozen 10-21) all retained as-is. Only two wording adjustments: the journal name in the review prompt, and replacing "wind energy / fluid / simulation paper" with "wind hazard / natural hazard paper" in the judge's invalid definition.
- Sampling rules finalized at 10-26 11:35 prior to viewing any candidates: .
- Results are presented alongside the original 433-unit results as the "NHESS subsample"; revising the overall conclusion would only be discussed if the direction were reversed.
- Completed at 10-26 12:2x: 10 papers (H1–H10), 314 ground-truth comment units, 60 model review reports (1,500 items), 20 blind evaluations; results placed at the top under "Latest: NHESS subsample results". Status reverted to Closed (negative result). 0 SU, 0 CU.

Key figures